

Resistance Value

Blue = 6

Grey = 8

Brown = $\times 10$

Gold = $\pm 5\%$

$\therefore 68 \times 10$

$= 680\Omega \pm 5\%$

The main reason for the difference in the calculated resistance and measured resistance is due to tolerance for example $\pm 5\%$ meaning the actual resistance can be slightly higher or lower than the expected value.